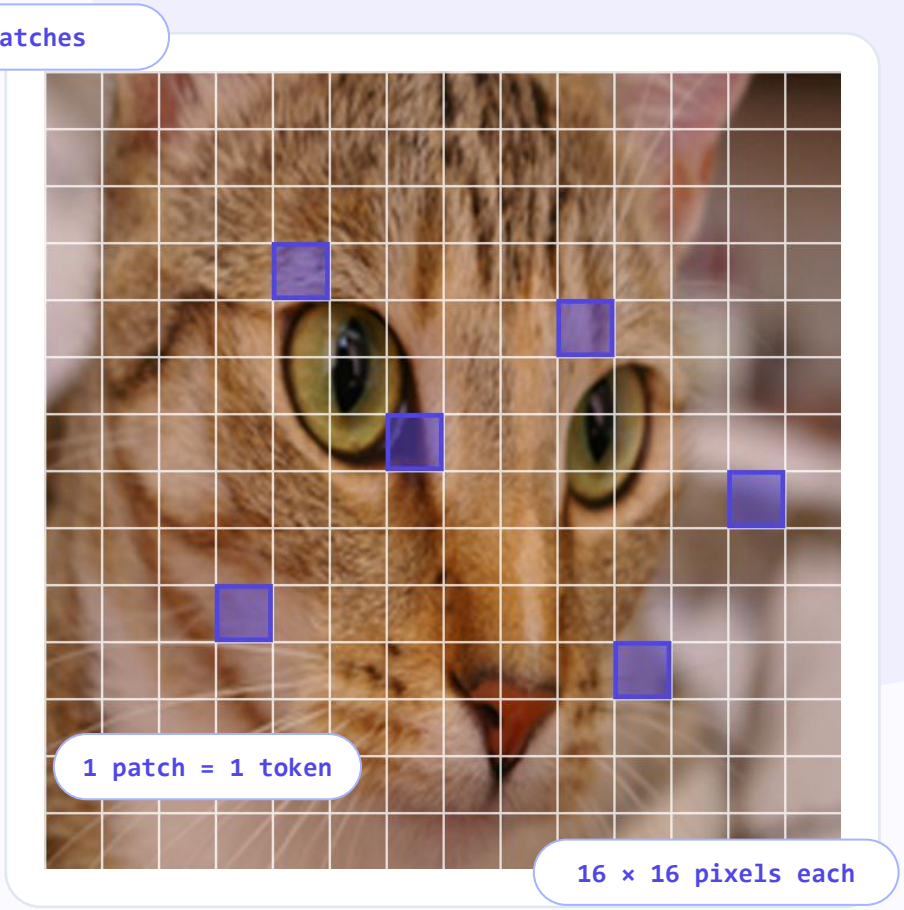
GENERATIVE AI · SEMINAR · BEYOND THE SYLLABUS

Vision Transformers: when an image becomes a sentence of patches

One idea from the paper “An Image is Worth 16x16 Words” (ICLR 2021): cut the picture into patches and let a Transformer read them.



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Subject: **Generative AI (26ITPE751)** Topic: **Vision Transformer (ViT)**

Vivekanand Education Society’s Institute of Technology · Department of Information Technology

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What we will cover

Eight short ideas, from why ViT exists to a live demo.

Why ViT? 01

CNNs look through small windows. ViT sees the whole picture.

Image to patches 02

A photo is cut into 16×16 pieces that act like words.

How ViT works 03

Six steps from pixels to a class label.

Self-attention 04

Every patch asks which other patches matter.

ViT vs CNN 05

Same accuracy, less compute, but it needs big data.

Demo 1 06

Patches and predictions with a pretrained ViT.

Demo 2 07

Where the model looks, and what a shuffle does.

Uses and limits 08

Where ViT is applied and what it still needs.

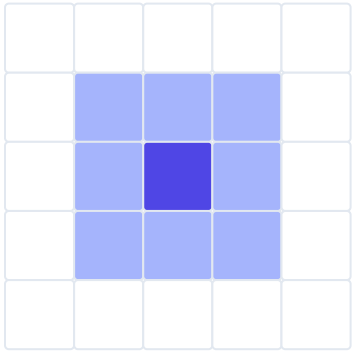
Source: Dosovitskiy et al., An Image is Worth 16x16 Words, ICLR 2021 (arXiv:2010.11929)

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CNNs see a little, ViT sees it all

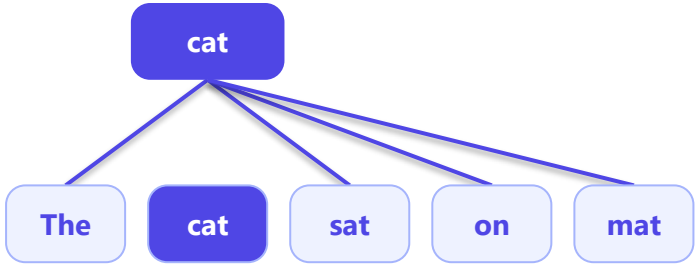
A CNN reads a photo through small windows. A Transformer lets every part talk to every other part.

CNN: small windows



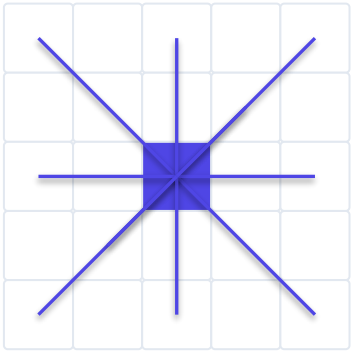
Each filter sees only a 3 × 3 neighbourhood. Distant parts meet only after many layers.

Transformer: everyone talks



Built for text. Every word is compared with every other word at the same time.

ViT: both ideas



Cut the picture into patches, then let every patch look at every other patch.

In one line: ViT takes the Transformer from text and uses it to read images.

Source: Dosovitskiy et al., An Image is Worth 16x16 Words, ICLR 2021 (arXiv:2010.11929) · diagrams are simplified illustrations

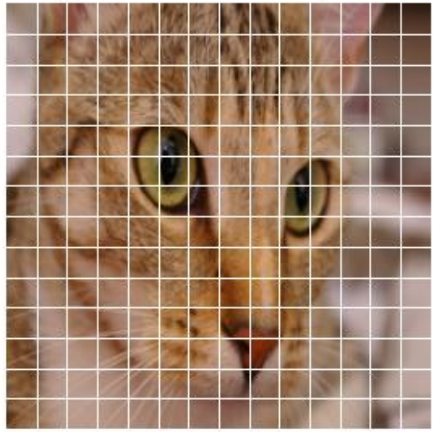
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An image becomes a sentence

ViT does not read pixel by pixel. It cuts the photo into 16 × 16 patches, and each patch plays the role of a word.

From our demo: real patches of the cat photo

Image cut into 14 x 14 patches



8 of the 196 patches (each one becomes a 'word')



PYTHON

```
patches = img.reshape(14, 16, 14, 16, 3)
patches = patches.swapaxes(1, 2).reshape(-1, 16, 16, 3)
patches.shape   # (196, 16, 16, 3)
```

196

patches per image (14 × 14)

768

numbers in one patch (16 × 16 × 3)

The sentence analogy

Sentence	→	**words**	→	**word vectors**
Image	→	**patches**	→	**patch vectors**

Source: Dosovitskiy et al., An Image is Worth 16x16 Words, ICLR 2021 (arXiv:2010.11929) · figure: our Colab demo (google/vit-base-patch16-224, scikit-image sample photo)

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How ViT works, step by step

Six steps from raw pixels to a class label such as “Egyptian cat”.

Cut into patches 01

224 × 224 image → 196 patches of 16 × 16 pixels.

Flatten and embed 02

Each patch becomes a vector of 768 numbers (a linear layer).

Add position 03

A position embedding tells the model where each patch was.

Add a [class] token 04

One extra token that gathers the answer for the whole image.

Transformer encoder × 12 05

Self-attention + MLP. All 197 tokens are processed together.

MLP head → label 06

The [class] token goes to a classifier that outputs the label.

ViT-Base 12 layers · hidden size 768 · 12 attention heads · 86 million parameters · 197 tokens (196 patches + [class])

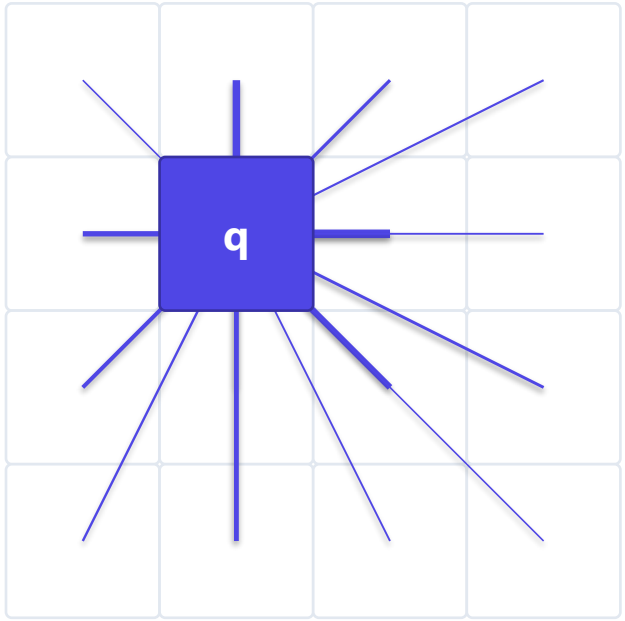
Source: Dosovitskiy et al., An Image is Worth 16x16 Words, ICLR 2021 (arXiv:2010.11929) · model sizes from Table 1 of the paper

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Self-attention in plain words

Every patch asks one question: which other patches are useful to understand me?

One query patch looks at all the others



Thicker line = higher attention score. The scores of q add up to 1.

1 **Ask**

Each patch asks: who is relevant to me?

2 **Score**

Every other patch gets a score. The scores add up to 1.

3 **Mix**

The patch collects information from others, weighted by the scores.

```
new value of q = sum over k of ( score(q, k) × value of k )
```

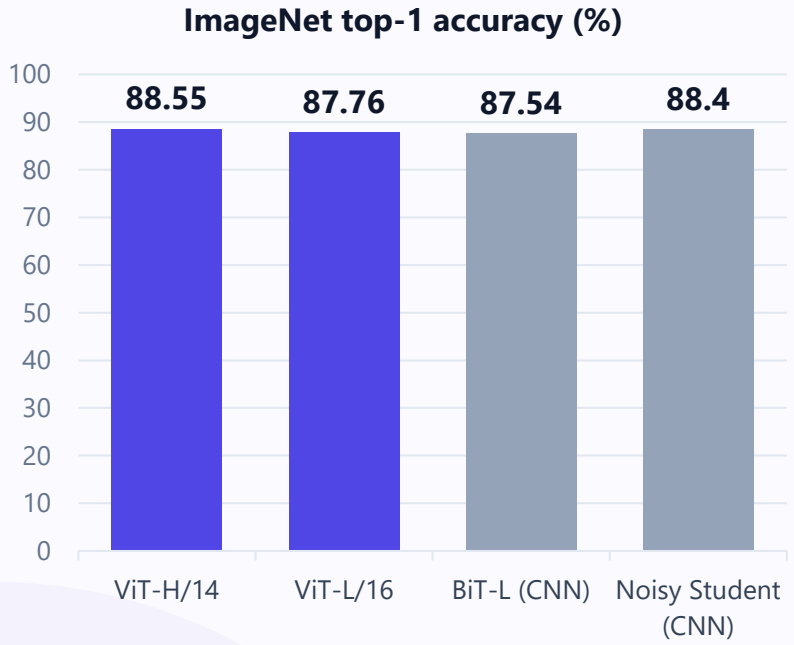
Why it matters: distant patches are linked in one layer.

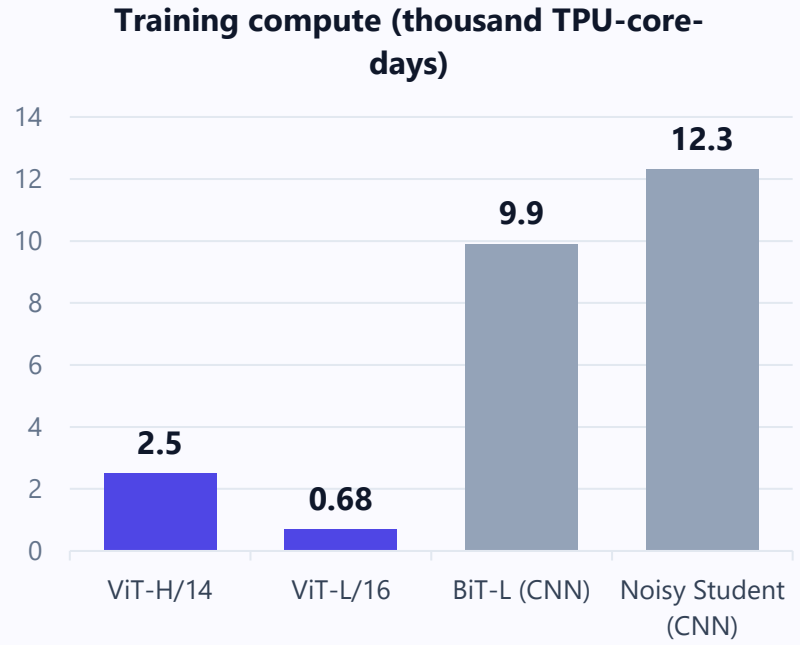
Source: Dosovitskiy et al., An Image is Worth 16x16 Words, ICLR 2021 (arXiv:2010.11929) · Vaswani et al., Attention Is All You Need (2017) · diagram is a simplified illustration

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Same accuracy, far less compute

On ImageNet the best ViT matches the best CNNs, using a fraction of the training compute.





indigo = Vision Transformer **slate** = CNN lower compute is better

Read it like this: ViT-L/16 needs about 15× less compute than BiT-L for similar accuracy (9.9k vs 0.68k).

Big data: ViT wins

After pre-training on 300 million images (JFT-300M), ViT matches or beats the best CNNs.

Small data: CNN wins

On ImageNet alone ViT is a bit below ResNets of similar size. CNNs have locality built in.

Hybrids

CvT and Swin Transformer mix convolution with attention to get both strengths.

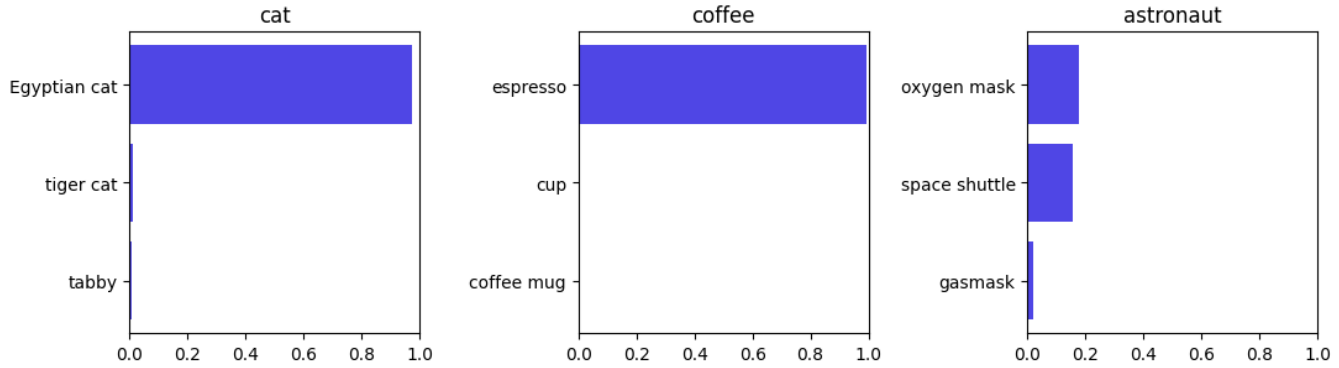
Source: Dosovitskiy et al., ICLR 2021, Table 2 (ImageNet top-1 after fine-tuning; TPUv3-core-days = pre-training compute). Noisy Student reports 88.4.

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Predictions with a pretrained ViT

We load google/vit-base-patch16-224 from Hugging Face and classify three photos. No training needed.

Top-3 predictions for each photo



PYTHON

```
inputs = processor(images=img, return_tensors="pt")
probs  = model(**inputs).logits.softmax(dim=-1)[0]
top3   = probs.topk(3)
```

98%

cat → Egyptian cat

99%

coffee → espresso

18%

astronaut → oxygen mask

ImageNet has no “astronaut” class, so confidence is low.

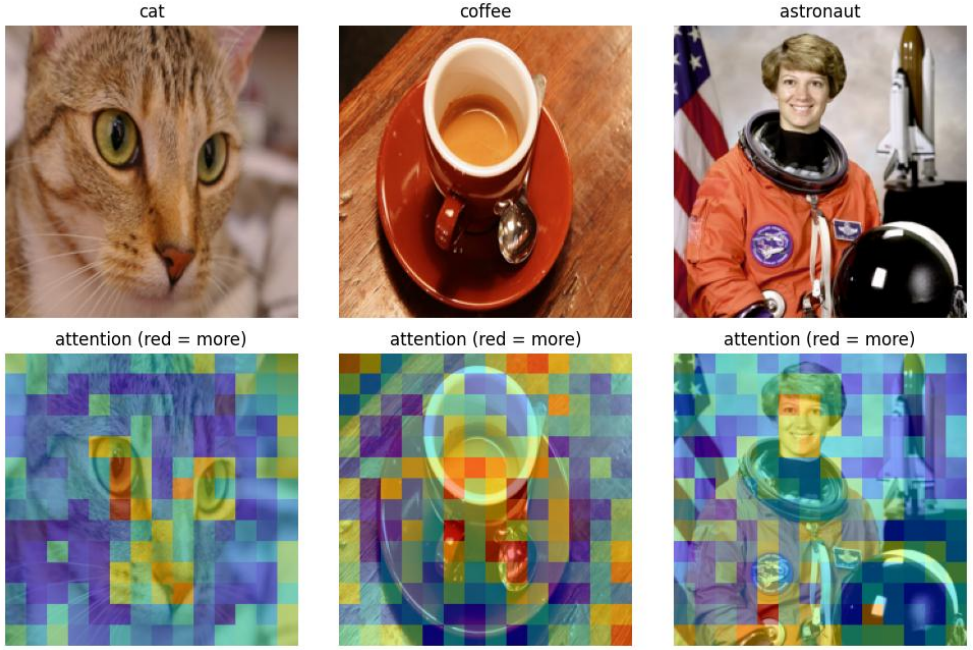
Source: our Colab demo (ViT_demo.ipynb): google/vit-base-patch16-224, scikit-image sample photos; ImageNet has 1000 classes

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Where does ViT look?

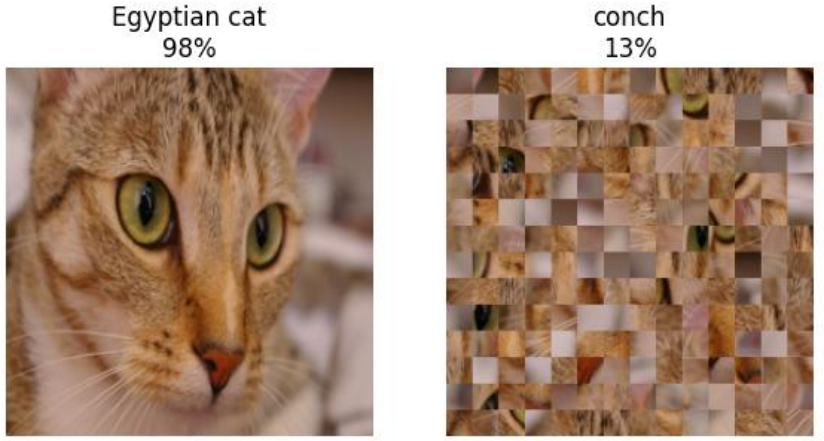
Attention shows where the [class] token looked. Shuffling the patches breaks the prediction.

Attention rollout (red = more attention)



Clearest for the cat's eyes; scattered for the coffee and astronaut photos.

Shuffle test: same patches, new order



Egyptian cat 98% → **conch 13%**

Same patches, new arrangement: the prediction breaks. This shows arrangement matters. We did not test a CNN or the position embedding, so it does not show why.

Source: our Colab demo (ViT_demo.ipynb) · attention rollout: Abnar & Zuidema (2020) · red = more attention

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Where ViT is used, and what it needs

ViT is now a standard building block for vision. It also has clear limits.

Image classification

Best ViT reaches 88.55% top-1 on ImageNet after large pre-training (paper).

Medical imaging

Reading X-ray and MRI images, for example to help detect tumours.

Self-driving

Understanding road scenes from camera images.

Multimodal AI

CLIP uses a ViT as its image encoder to link pictures with text.

Limits needs very large pre-training data (JFT-300M = 300M images) · CNNs are better on small datasets · training still costs thousands of TPU-core-days

Source: Dosovitskiy et al., An Image is Worth 16x16 Words, ICLR 2021 (arXiv:2010.11929) · Radford et al., CLIP (ICML 2021) · application areas are typical examples, not results of this paper

CONCLUSION

Thank you

One idea, six steps: an image is a sentence of patches, and a Transformer can read it.

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Generative AI (26ITPE751) · Beyond-syllabus seminar

Questions?

PATCHES

An image is a sentence

16×16 patches act as words.

ARRANGEMENT

Patch layout matters

Shuffled patches: 98% → 13%.

ATTENTION

Global in one layer

Every patch sees every patch.

EFFICIENCY

Same accuracy, less compute

ViT-L/16: ~15× less than BiT-L.

DATA

Needs a lot of it

CNNs win on small datasets.

IMPACT

A vision building block

Also inside CLIP.

Key reference A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, N. Houlsby. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. ICLR 2021. https://arxiv.org/abs/2010.11929

Also: Vaswani et al., Attention Is All You Need (2017), arxiv.org/abs/1706.03762 · Abnar & Zuidema, Attention Rollout (2020), arxiv.org/abs/2005.00928 · Code: github.com/google-research/vision_transformer